Oscillations

Introduction

Oscillations is the repetitive motion back and forth about equilibrium position.

— For an oscillatory motion:

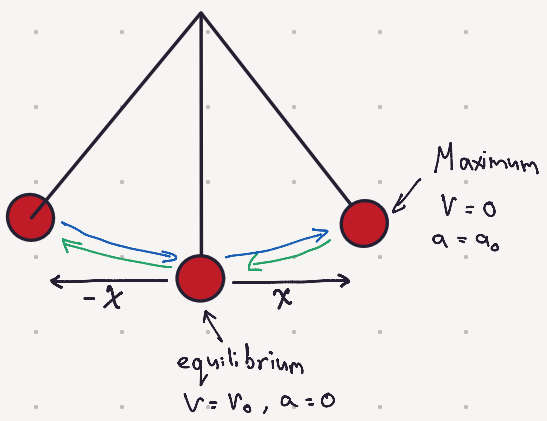
- The restoring force always act towards equilibrium position → acceleration act towards equilibrium
- At equilibrium → acceleration = 0
  → Velocity = $V_{max} = V_0$
- At max displacement (x_0) → acceleration = a_0
  → Velocity = 0

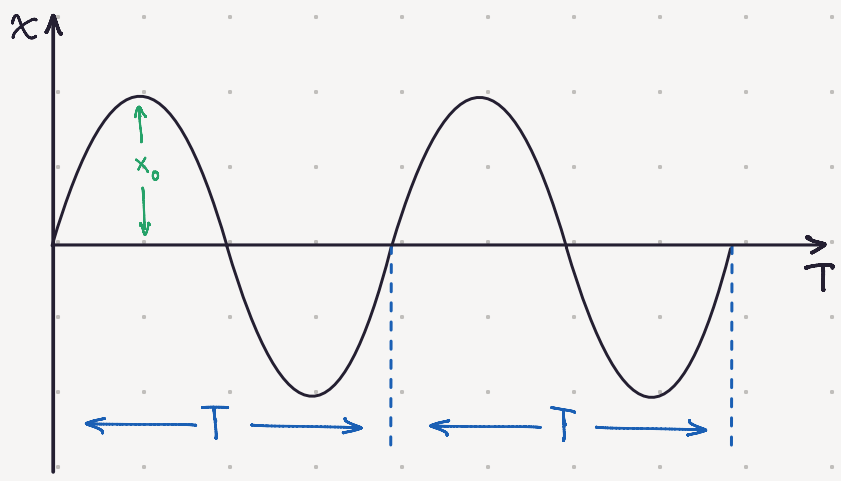
Key Definitions

- Equilibrium: Position where net force on object = 0
- Displacement (x): distance from a point in a given direction
- Amplitude (x_0): Maximum displacement
- Period (T): time for one complete oscillation
- Frequency (f): Number of oscillations per unit time

$$f = \frac{1}{T} \text{ or } T = \frac{1}{f}$$

E.g.





Simple Harmonic Motion

Simple harmonic motion is a special type of oscillation where:

- Acceleration is directly proportional to displacement, but always in the opposite direction (towards equilibrium)

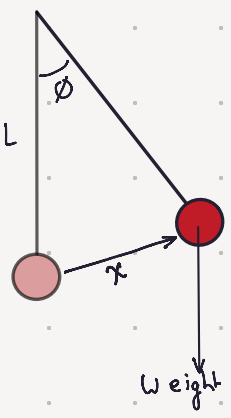
— Symbols: $a \propto -x$ (proportional; opposite)

— Equation: $a = -\omega^2 x$ (ω^2: constant)

Angular frequency (ω)

- $\omega = 2\pi f = \frac{2\pi}{T}$
- Unit: rad s^{-1}

E.g. Pendulum

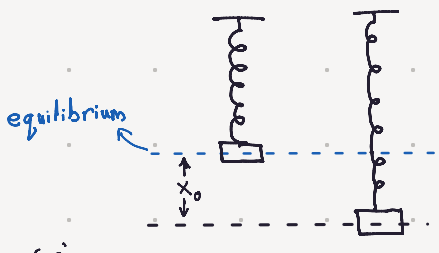


For small Ø, $\sin Ø = \frac{-x}{L}$

$F = ma = mg \sin Ø$

$a = -\frac{g}{L} x \rightarrow \omega^2 = \frac{g}{L}$

E.g. Springs



spring constant

$F_{net} = -k\Delta x = -k x_0 \rightarrow F = ma$

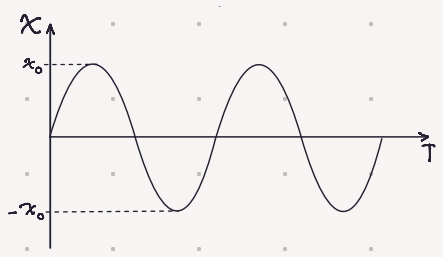
$\rightarrow ma = -kx \rightarrow a = -\frac{k}{m} x$

So, $\omega^2 = \frac{k}{m}$

SHM Equations and Graphs

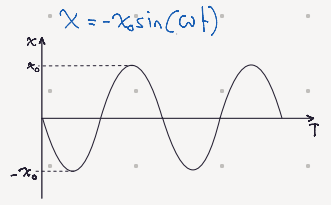
Displacement

① If $t = 0$ is equilibrium, use: $X = X_0 \sin(\omega t)$

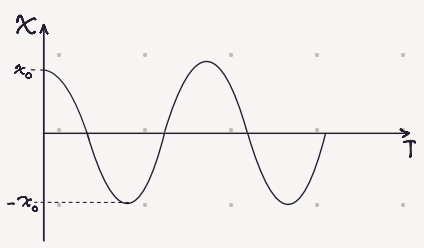


Moves in the -ive direction at first

$X = -X_0 \sin(\omega t)$



② If $t = 0$ is a maxima, use: $X = X_0 \cos(\omega t)$



③ If $t = 0$ is elsewhere, use general:

$X = X_0 \sin(\omega t + \phi)$

Velocity

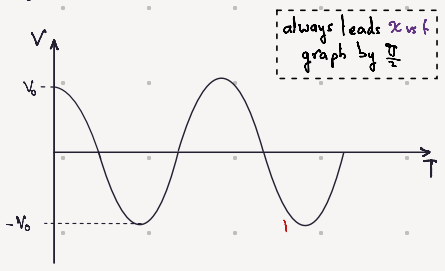
By differentiating the displacement, we get:

$V = V_0 \cos(\omega t)$

or

$V = \pm \omega \sqrt{x_0^2 - x^2}$

This assumes $t = 0$ is equilibrium and moving in +x



Velocity is max at $x = 0$, so: $V_{max} = V_0 = \omega X_0$

Acceleration

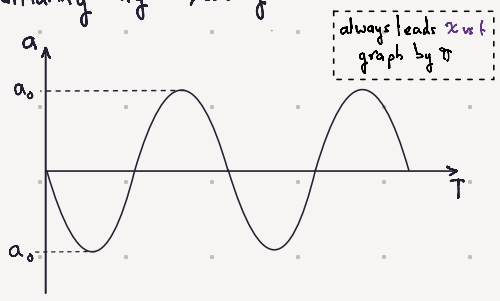
By differentiating again, we get:

$a = -a_0 \sin(\omega t)$

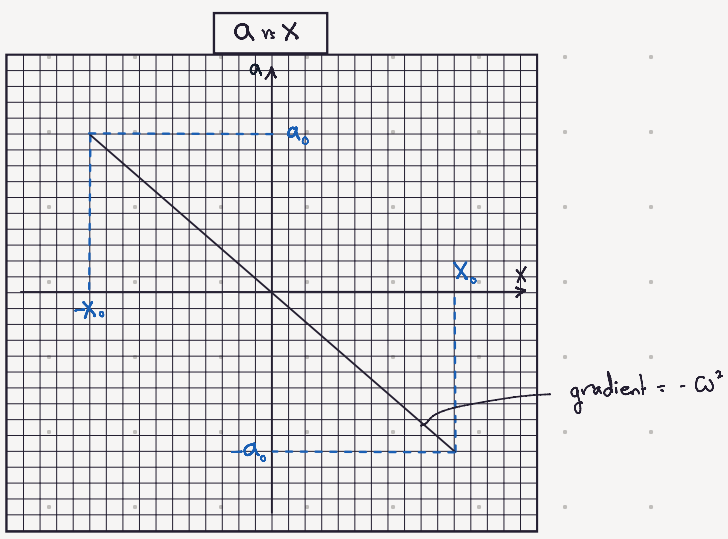
or

$a = -\omega^2 x$

This assumes $t = 0$ is equilibrium and moving in +x



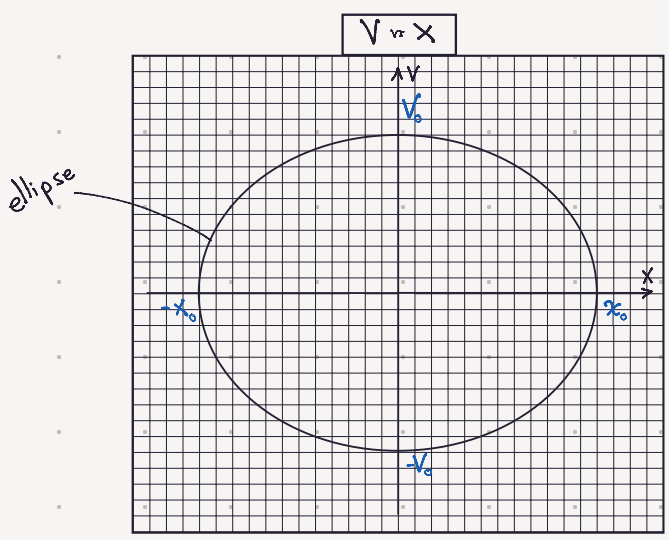
Acceleration is max at x_0, so: $a_{max} = a_0 = \omega^2 x$



Features:

- A straight line through origin → acceleration is proportional to displacement
- Negative gradient → acceleration and displacement are always in opposite directions

} Indication of SHM



Features:

- X-intercept → $x = x_0$ → $V = 0$ (momentarily at rest)
- Y-intercept → $x = 0$ → $V = V_0$

Energy in a SHM

In a simple harmonic motion (no damping), the energy continuously changes between kenetic and potential

Total energy

$$E_T = E_p + E_k = \frac{1}{2} m\omega^2 x^2 + \frac{1}{2} m\omega^2 (x_0^2 - x)$$

$$\boxed{E_T = \frac{1}{2} m\omega^2 x_0^2}$$

Features:

- Total energy is constant
- $E_T = E_{p,Max} = E_{k,Max}$
- Proprotional to x^2
  double amplitude → quadruple the energy

Potential energy

$$E_p = \frac{1}{2} k x^2 \xrightarrow{\omega^2 = \frac{k}{m}} \boxed{E_p = \frac{1}{2} m\omega^2 x^2}$$

At equibrium: $E_p = 0$

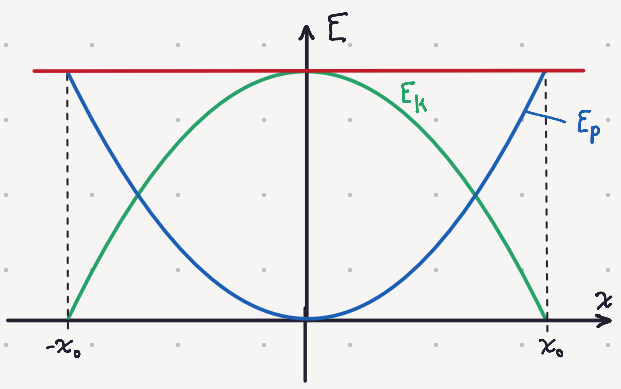
At maximas: $E_p = Max$

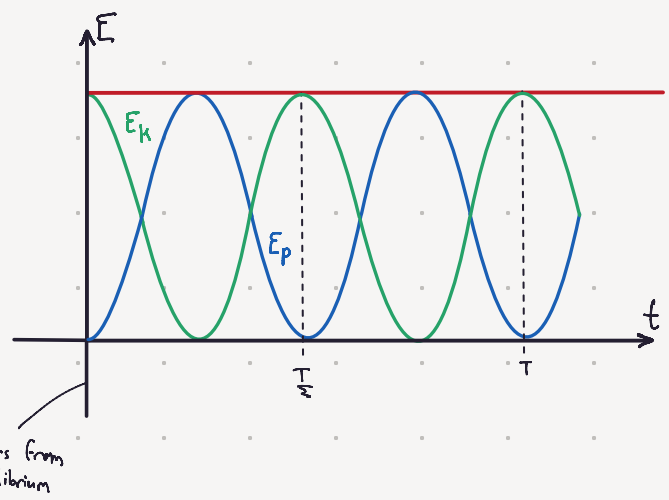
Kenetic energy

$$E_k = \frac{1}{2} mv^2 \xrightarrow{v = \pm\omega\sqrt{x_0^2 - x^2}} \boxed{E_k = \frac{1}{2} m\omega^2 (x_0^2 - x)}$$

At equibrium: $E_k = Max$

At maximas: $E_k = 0$





Damping

In reality, there is friction and resistance that oppose motion meaning that energy doesn't stay constant, this is called damping.

Damping: The loss of energy - hence decrease in amplitude - due to resistive forces.

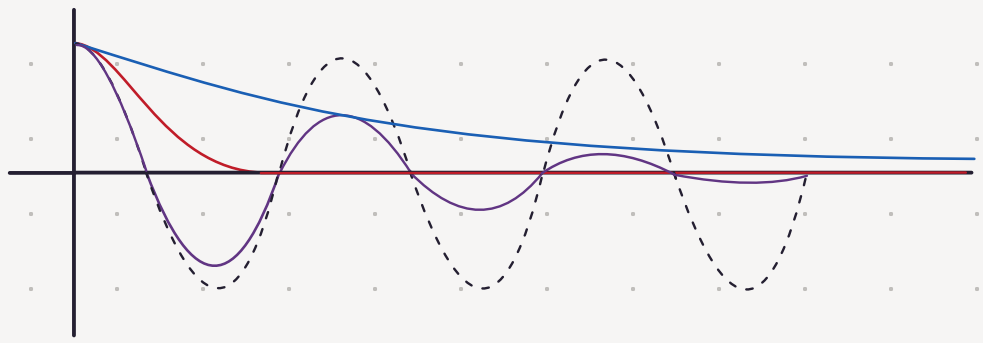
Note: the affect of damping on period is negligible

Types of damping

Light damping: Oscillations continues and the amplitude decreases gradually

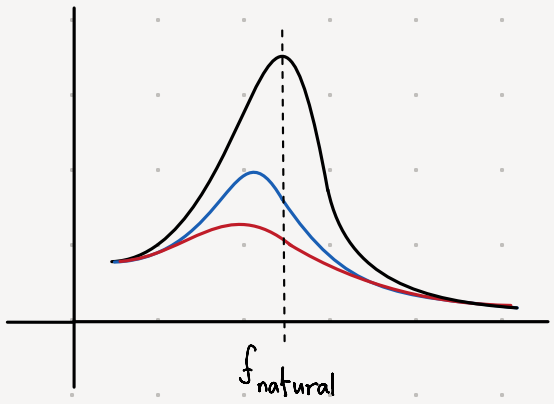
Critical damping: Exact damping that prevents any oscillation → returns to equilibrium in the shortest possible time.

Heavy damping: Resistive force > Critical value
→ no oscillation but takes more time than critical.



Forced Oscillation

When an object oscillates on its own, with no energy lose or gain, it is free oscillating at its natural frequency.
However, forced oscillation occurs when an external periodic force is applied causing the object to vibrate at the frequency of that force.



Affect of damping on resonance:

① lower peak – decreases max amplitude

② flatter peak

③ Peak shifts to the left – max amplitude at a lower frequency

Resonance is destructive whenever a structure or object is forced to vibrate at its natural frequency, because energy is continually transferred to the oscillation and the amplitude builds up until the object fails — so a structure can vibrate itself apart or crack, glass can shatter, and machine or vehicle parts can vibrate violently at particular driving frequencies. The general way to prevent this is to add damping: a resistive force dissipates the energy of the oscillation, so the resonance peak becomes flatter and the amplitude at resonance is much smaller

Resonance

Definition:

When the driving frequency - due to force - matches the natural frequency the oscillations is at maximum amplitude.

→ At resonance energy transfer is most effecient

Applications:

- Quartz crystals to produce ultrasound
- Quartz crystals in watches to keep timing
- NMR/MRI scanning, resonance of nuclei used to build image
- Microwave ovens, resonance of water mol.